MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

(2024 ADMISSION ONWARDS)

MG2DSCNBC100 – BIOCHEMISTRY OF ENERGY TRANSFORMATION

Duration: 2hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Each question is tagged with the appropriate Bloom's Revised Taxonomy level (Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.) and the relevant course outcome (CO) number.
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S).

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Give an example of Transferases [K] [3]
   a) Transaminase b) Lipase c) Oxidase d) Isomerase
2. Which of the following components primarily make up the cell membrane? [K] [1]
   a) Proteins & CHOb) Proteins & Lipidsc) Lipids &CHO d) CHO & Nucleic acids
3. Name the nitrogenous base present in RNA but absent in DNA [U] [3]
   a)Thymine b) Uracil c) Cytosine d) Adenine
4. Which type of carrier transports multiple molecules in both directions across the membrane? [K] [1]
   a)Uniport b) Symport c) Antiport d) Channel proteins
5. ……is a process in which RNA is synthesized from DNA. [U] [3]
   a) Translationb) Transcriptionc) Replication d) Phosphorylation
6. Transport that doesnot require energy is ……. [K] [1]
   a) Active transportb) Endocytosis c) Passive transport d) Vesicular transport

[1 X 6 = 6]

Part B

Short Answer Type Questions

Answer **2** Questions

Each question carries **2**marks

7. Define osmosis [K] [1]
8. Discuss the classification of high energy compounds. [U] [2]
9. Define Replication [K] [3]
10. Explain the IUB classification of enzymes. [U] [3]

[2 X 2 = 4]

Part C

Short Essay Type Questions

Answer **4** Questions

Each question carries **5** marks

11. Explain the components of ETC[U] [2]
12. Explain DNA recombination technology. [U] [3]
13. Discuss the Watson crick model of DNA.[U] [3]
14. Explain the ping pong mechanism of facilitated diffusion?[U] [1]
15. Discuss on carrier proteins.[U] [1]
16. Compare active and passive transport.[U] [1]

[5 X 4 =20]

Part D

Essay Type Questions

Answer **2** Questions

Each question carries **10** marks

17. Recall the factors affecting diffusion rate. [K] [1]
18. Explain Electron Transport Chain. [U] [2]
19. Explain Protein Biosynthesis. [U] [3]
20. Explain the different types of transport systems. [U][1]

[10 X 2 =20]